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AI Health message intervention: The role of message customization and message source in breast cancer screening among women of color

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ABSTRACT

Objectives: To examine the effectiveness of breast cancer screening messages with varying levels of customization (generic, targeted, and tailored) and to compare AI-generated versus human-generated messages.
Methods: A between-subjects experimental design with a control condition was employed. Message content followed a standardized structure and varied by level of customization: generic, targeted (demographic-based), and tailored (perceived susceptibility- and barrier-based). Messages were developed by either the authors or GenAI (ChatGPT-4o). A total of 391 participants recruited via Prolific were randomly assigned to five groups (generic, targeted-human, targeted-AI, tailored-human, and tailored-AI). Self-efficacy, behavioral intentions, attitudes, and message believability were measured using different scales.
Results: Customized (tailoring and targeting) health messages performed comparably to generic messages in shaping positive health outcomes. GenAI-generated messages also produced outcomes comparable to those of human-generated messages under standardized conditions. Significant negative indirect effects through message believability for the human-tailored condition was found relative to the generic condition.
Conclusions: GenAI may be a useful tool for developing and customizing scalable health messages. Its effectiveness depends not only on customization but also on maintaining message quality, including readability, clarity, coherence, naturalness, and credibility.
Practical implications: GenAI may support health practitioners in developing customized and scalable breast cancer messages. However, professional review remains necessary to ensure that the message is culturally appropriate, responsive to patient concerns, and suitable for use alongside patient-provider communication.

1. Introduction

Breast cancer remains a leading cause of cancer-related death among women in the United States [1], with approximately 300,000 new cases and 43,000 deaths reported in 2023 [2]. Despite advances in medical care, substantial racial and ethnic disparities persist [3]. Black women experience a breast cancer mortality rate that is 38% higher than that of White women [4]. Disparities are also evident among Hispanic women. National survey data indicate that approximately 60% of Hispanic women receive recommended breast cancer screening, compared with 68% of White women [2]. These disparities may be partly attributable to negative attitudes toward screening, including fear of pain, concerns

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about radiation exposure, psychological barriers, and financial constraints, which can contribute to delayed or forgone screening and treatment [5].

To address disparities in breast cancer screening, healthcare practitioners have developed various strategies to improve awareness and encourage screening behaviors [6]. One such strategy is message customization. Specifically, message customization is conceptualized along a continuum ranging from generic, one-size-fits-all messages to highly tailored messages [7]. Generic messages are broad, standardized forms of information designed for mass audiences, whereas targeted messages adapt content based on observable group characteristics, such as age, gender, or ethnicity [7]. At the highest level, tailored messages incorporate individual sociopsychological attributes, including health beliefs, efficacy expectations, and motivations, to personalize message content for each individual [8]. As a result, customized messages (i.e., targeted and tailored messages) align health information with audience-specific characteristics, needs, and preferences, making health messages more relevant and easier to understand [7,8]. For example, delivering breast cancer screening information through pamphlets and television can improve engagement with screening information among Black individuals [9]. Beyond delivering messages through appropriate channels, health practitioners also tailor messages to individuals' beliefs, prior experiences, and concerns [10]. Such customized messages are more likely to be perceived as personally relevant because they help individuals better understand screening recommendations and make informed decisions [10].

However, existing research has predominantly focused on White participants, with comparatively limited attention to racially minoritized populations [11–15]. As a result, health practitioners have limited evidence on which to base effective strategies for women of color, which may further exacerbate disparities in breast cancer screening. Second, research targeting people of color has largely relied on geographic, cultural, behavioral, and demographic characteristics, such as lived experiences, language differences, and family history of breast cancer [11–15]. Although these factors are empirically valuable, such approaches may remain relatively surface-level and limit the generalizability of health materials across diverse populations and health contexts. To develop more generalizable health materials, this study applies the Health Belief Model (HBM) as the guiding framework for message design. Breast cancer screening messages are a form of patient education because they help individuals interpret risk information, understand screening recommendations, and make informed decisions about preventive care. For women of color, effective patient education must also address structural and psychological barriers that may shape screening decisions. The present study therefore examines whether demographic targeting, Health Belief Model-based psychological tailoring, and GenAI-assisted message generation can improve the relevance and persuasive impact of breast cancer patient education.

The HBM posits that preventive health behaviors are influenced by how individuals perceive health risks (i.e., susceptibility and severity) and the benefits and barriers associated with taking action [6,16]. Specifically, perceived susceptibility and perceived severity capture how individuals perceive the threat of a health condition, whereas perceived benefits, perceived barriers, and self-efficacy reflect how they evaluate the recommended health behavior [16]. We incorporate perceived susceptibility and perceived barriers as the theoretical foundation for message design because these constructs directly address major factors contributing to delayed or forgone breast cancer screening, including low perceived risk and concerns about screening barriers [5]. In this case, message tailoring is designed by aligning message content with individuals' levels of perceived susceptibility (e.g., personally relevant risk) and addressing their perceived barriers (e.g., concerns about cost, fear, or accessibility). Because tailored messages directly engage individuals' specific psychological characteristics, they are theoretically expected, and have been shown, to be more effective than generic or demographically targeted messages by enhancing the perceived personal relevance of health information [17–20]. Consistent with this evidence, tailored breast cancer screening messages have been found to increase patients' engagement with mammography [21]. However, research focusing specifically on women of color remains limited [15]. Given the HBM's proposition that preventive behaviors are driven by individuals' psychological factors, we hypothesize:

H1. : Tailored messages will be more effective than targeted and generic messages in promoting screening behavior and attitudes.

Existing studies have often examined message tailoring and GenAI-assisted message generation separately [20,22]. The present study integrates these two dimensions by comparing generic, demographically targeted, and Health Belief Model-based psychologically tailored messages developed by human authors or GenAI among Black and Hispanic women. The contribution therefore extends beyond testing the established expectation that tailored messages may outperform generic information [23]. By simultaneously varying customization depth and message source, the study evaluates whether GenAI can support theoretically informed patient education at different levels of customization and whether psychological tailoring provides incremental value beyond demographic targeting.

To promote healthy behaviors, health practitioners have traditionally relied on messages developed by professional health communicators [24]. However, developing customized messages is often time- and resource-intensive [25]. Accordingly, generative artificial intelligence (GenAI) offers them a promising tool for developing health materials more efficiently. Rather than relying solely on professionals, practitioners can use GenAI to rapidly generate health messages that are personalized to specific audiences and communication goals. By providing prompts that specify audience characteristics, communication objectives, and relevant health information, practitioners can rapidly get messages that are generated, revised, and refined by GenAI [24–26]. This process enables the efficient development of highly customized health messages, and at the same time, reduces the necessary time and resources. However, whether GenAI-generated messages are as effective as those written by professional communicators remains unclear. Existing studies have reported mixed findings. The reason is that message effectiveness varies according to different factors, such as message quality, audience characteristics, and the degree of personalization [27,28]. Although emerging evidence suggests that GenAI can generate culturally appropriate and coherent health messages [29], little is known about whether customized messages improve breast cancer education and screening promotion among women of color. Addressing this question is important for determining whether GenAI can design effective health messages at scale. Therefore, we ask:

RQ1. : How do AI-generated messages compare to human-generated messages in their effects on screening behavior and attitudes?

In addition, understanding why customized health messages are effective is important because identifying the underlying mechanisms can help practitioners better understand how messages influence audiences [21]. Accordingly, this study examines self-efficacy and message believability as key mechanisms. Self-efficacy refers to individuals' confidence in their ability to perform a recommended health behavior, such as obtaining a mammogram [6,16]. Guided by the HBM and Social Cognitive Theory [8,30], customized messages align health information with the characteristics and needs of their intended audiences, thereby increasing individuals' confidence in obtaining a mammogram. In contrast, message believability reflects the extent to which individuals perceive health information as accurate and trustworthy [31]. GenAI- and human-generated messages may differ in perceived believability because their content may contain cues that influence perceptions of trustworthiness [32]. Thus, message source may influence how believable audiences perceive a message to be. Greater message believability, in turn, may increase acceptance of screening recommendations. Accordingly, we propose the following hypothesis and research

question.

H2. : Self-efficacy will mediate the relationship between message condition and screening attitudes and behavioral intention.

RQ2. : Does message believability mediate the relationship between message condition and screening attitudes and behavioral intention?

2. Methods

2.1. Participants

Upon IRB approval from the lead author's institution, we conducted an online between-subjects experiment in December 2024 using Prolific, a widely used platform that enables researchers to recruit participants with diverse demographic characteristics. Participants were randomly assigned to one of five conditions: a generic message condition or one of four customized message conditions crossing message source (GenAI vs. human) and customization level (targeted vs. tailored). To focus on women of color, recruitment was limited to Black and Hispanic women in the U.S., aged 18–60. Although routine screening guidelines typically begin at age 40, younger women were included to capture early-stage awareness and intention formation, particularly given the increasing risk of early-onset breast cancer among women of color [2,33]. Of the 467 initial respondents, only those who finished the full survey and passed two attention checks were retained. This resulted in 391 valid responses, meeting the required sample size of 305 participants to detect a small effect size ($f = 0.20$) with 80% power at $\alpha = 0.05$. See Table 1 for demographic characteristics of participants.

2.2. Intervention

Participants were randomly assigned to receive one of the experimental health messages. To ensure consistency across conditions, all intervention messages followed the same overall structure. Specifically, each message starts with background information, followed by age- and race-specific information, perceived susceptibility- and barrier-related content, and a concluding recommendation for breast cancer screening. Only the sections relevant to age, race, susceptibility and barrier were modified, while the rest of parts (i.e., background information, concluding recommendation) remained consistent across conditions. Each message was approximately 400 words in length. The complete intervention stimuli are provided in Appendix A. The following sections describe how intervention messages are designed by chunks.

First, guided by the definition of targeted and tailored messages, targeted messages address information on individuals' age and race, whereas tailored messages addressed individuals' levels of perceived susceptibility and perceived barriers. Age- and race-specific information was obtained from the Centers for Disease Control and Prevention (CDC) [33]. Levels of susceptibility and barriers were self-reported by participants. In the targeted condition, participants received messages customized to their age and racial group. For example, Black women aged 18–39 received messages emphasizing that younger women remain at risk of breast cancer, supported by age- and race-specific data. In contrast, Hispanic women aged 40 years and older received messages highlighting the increasing risk of breast cancer in their ages, together with statistics specific to Hispanic women. A similar approach was applied to other age and racial groups.

Second, guided by definitions of perceived susceptibility and perceived barriers, tailored messages were designed to match participants' psychological profiles. Specifically, participants with low perceived susceptibility received messages emphasizing that breast cancer can develop even in the absence of noticeable symptoms and highlighting the importance of remaining vigilant. In contrast, participants with high perceived susceptibility received messages acknowledging their elevated risk of breast cancer and encouraging proactive screening behaviors. A similar approach was applied to perceived barriers, with message content adapted to address participants' self-reported level of barriers to obtaining a mammogram.

Finally, human-generated messages were developed by the authors, who have training in health communication and public health. The messages were subsequently reviewed and refined to ensure clarity, consistency, and appropriateness. In contrast, GenAI-generated messages were produced using ChatGPT−4o with standardized prompts (Appendix B). The prompts were designed to ensure consistency in message structure, length, and informational content while allowing the

Table 1
Demographic characteristics of participants.

Characteristics	Frequency	Percentage
Message Customization		
Generic	128	32.7%
Targeted	133	34.0%
Tailored	130	33.2%
Message Source		
AI	195	49.9%
Human	196	50.1%
Groups		
Generic	127	32.5%
Human-Targeted	73	15.3%
AI-Targeted	60	18.7%
Human-Tailored	60	15.3%
AI-Tailored	71	18.2%
Ethnicity		
Black	199	50.9%
Hispanic	192	49.1%
Age		
18–39	198	50.6%
40 and above	193	49.4%
Diagnosed Cancer*		
Diabetes	47	12.0%
Infections	79	20.2%
Injuries and Wounds	111	28.4%
Environmental Health	28	7.2%
Other	95	24.3%
None/Rather not say	155	39.6%
Screening History*		
Mammogram	127	32.5%
Breast ultrasound	48	12.3%
Clinical breast exam by a healthcare provider	103	26.3%
Other	3	0.8%
I have not had any breast cancer screenings in the past 12 months	183	46.8%
Perceived susceptibility		
Yes	93	23.8%
No	173	44.2%
Not sure	125	32.0%
Perceived barrier		
Yes	65	16.6%
No	282	72.1%
Not sure	44	11.3%
Education		
Less than High School	4	1.0%
High School/GED	46	11.8%
Some College	87	22.3%
2-year College Degree	44	11.3%
4-year College Degree	151	38.6%
Master's Degree	48	12.3%
Doctoral Degree	5	1.3%
Professional Degree (JD, MD)	6	1.5%
Income		
Below $15,000	25	6.4%
$15,000 - $24,999	23	5.9%
$25,000 - $34,999	40	10.2%
$35,000 - $49,999	48	12.3%
$50.000 - $74,999	71	18.2%
$75,000 - $99,999	86	22.0%
$100,000 - $149,999	66	16.9%
$150,000 or more	32	8.2%
Overall	391	

* : Participants could select all applicable responses.

content to be customized according to the experimental conditions.

2.3. Design and setting

The survey was administered via Qualtrics and began with pre-screening questions. Participants first reported their race, age, and health-related background information (e.g., prior cancer diagnosis and previous breast cancer screening behaviors). They also self-reported the levels of susceptibility to breast cancer and perceived barriers to obtaining a mammogram. Based on their responses, participants were classified into high or low levels for each characteristics. Specifically, a response of "yes" was categorized as high, whereas responses of "no" or "not sure" were categorized as low.

Based on these classifications, participants were randomly assigned to one of the five experimental conditions. Within the targeted conditions, participants received messages customized according to their age (18–39 vs. 40 years and older) and race (Black vs. Hispanic). Within the tailored conditions, message customization further incorporated participants' self-reported levels of perceived susceptibility and perceived barriers (high vs. low). Participants assigned to the generic condition received a standard message without customization.

After randomization, participants were instructed to read the intervention for two minutes. They then completed measures of self-efficacy and message believability, followed by assessments of attitudes toward breast cancer screening and behavioral intentions. Finally, participants reported additional sociodemographic information (e.g., education and income). The survey took approximately 7 min to complete, and participants received $1.12 in compensation (equivalent to an hourly rate of $9.60).

2.4. Measurement

Message believability (4 items) [34] and attitude (3 items) [35] were measured both using semantic differential scales. The following constructs were measured using Likert-type scales (1 = "Strongly disagree" to 5 = "Strongly agree"). Perceived barrier was measured with 9 items [36], perceived susceptibility was measured with 3 items [35], and self-efficacy used 3 items [35]. Behavior intention measured with 3 items [35]. Measurement items, reliability, means, and standard deviations of the measurement variables were presented in Table 2.

3. Results

To answer H1, a univariate analysis of variance with SPSS 29.0 using 5000 bootstrapped samples and 95% bias-adjusted confidence intervals was conducted to test the effect of customization level on dependent variables: attitudes and behavioral intention. Results showed the effect of customization level on attitude was not statistically significant, F (2, 388) = .59, p = .55, partial η^2 = .003. Similarly, the effect on behavioral intention was neither statistically significant, F (2, 388) = 1.54, p = .21, partial η^2 = .01. H1 was not supported.

To answer RQ1, a similar analysis was conducted to test the effect of message source. Results showed that the effect on attitude was not statistically significant, F (1, 389) = .013, p = .91, partial η^2 = .001. Similarly, the effect on behavioral intention was not significant, F (1, 389) = 1.77, p = .18, partial η^2 = .005.

Finally, to answer H2 and RQ2, a mediation model was run with the PROCESS model 4 using 5000 bootstrapped samples. No significant indirect effect through self-efficacy was observed for attitudes or behavioral intention, as all hypothesized indirect effects had confidence intervals that included zero. Therefore, H2 was not supported. However, significant negative indirect effects through message believability were observed for the human-tailored condition relative to the generic condition. Specifically, the human-tailored condition was indirectly associated with less favorable attitudes (-.11, 95% CI [–.20, –.028]) and lower behavioral intentions (-.14, 95% CI [–.28, –.04]).

Table 2
Measurement items, reliability, means, and standard deviations (SD).

Variables	Mean	SD	Inter-item Reliability
Perceived susceptibility	3.25	1.01	.83
It is likely that I will get breast cancer.			
I'm at the risk of getting breast cancer.			
I am concerned about the likelihood of developing breast cancer in the future.			
Perceived barriers	1.73	.70	.82
I am afraid of getting screened for breast cancer because I might find out something is wrong.			
I am afraid to get screened for breast cancer because I don't understand what will be done.			
I don't know how to go about getting screened for breast cancer.			
Getting screened for breast cancer would be embarrassing.			
Getting screened for breast cancer would be time consuming.			
Getting screened for breast cancer would be painful.			
I cannot remember to schedule a screening for breast cancer.			
I have other problems more important than getting screened for breast cancer.			
I am too old/young to need a routine screening for breast cancer.			
Self-efficacy	3.97	.84	.80
I am able to get screened to prevent contracting breast cancer.			
I can easily get screened to protect myself from breast cancer.			
I have the skill, time and money to get screened for breast cancer.			
Message believability	4.57	.61	.92
- How much do you think the message you just read is …			
Not believable – Believable			
Inaccurate – Accurate			
Inauthentic – Authentic			
Fake – Realistic			
Behavioral Intentions	4.01	1.12	97
Get screened for the breast cancer.			
Try to receive screening for breast cancer such as a mammogram.			
Actually take a screening for breast cancer.			
Attitude	4.59	.61	.82
- To you, breast cancer screening such as self-exam and mammograms, feels …			
Bad– Good			
Foolish – Smart			
Unfavorable – Favorable			

4. Discussion and conclusions

4.1. Discussion

Contrary to our expectation, message believability may suppress the persuasive effectiveness of the human-tailored condition. One possible explanation is that the generic message was developed by health professionals at the CDC and may therefore have been more readable, coherent, and stylistically polished. By contrast, the human-tailored message incorporated additional elements based on participants' demographic characteristics. These customized elements may not have been integrated as smoothly into the overall message, potentially reducing its coherence and naturalness. Participants may feel abrupt transitions between tailored sections or perceived some information as insufficiently connected to their personal circumstances. Such perceptions may have weakened message believability and, consequently, reduced the message's persuasive impact. This finding suggests that effective tailoring requires the integration of customized content into a natural and professionally written narratives.

Similarly, contrary to the expected advantage of tailoring, no statistically significant differences were detected by customization level. At the same time, GenAI-generated messages produced observed attitudes and behavioral intentions similar to those associated with human-developed messages under standardized conditions. The absence of a detectable disadvantage provides initial practical evidence that GenAI can be useful for creating theoretically guided, customized patient-education materials when prompts are standardized and the content is reviewed by health communication professionals. The practical value of GenAI may therefore lie less in outperforming human writers than in supporting the scalable production of customized education without a detectable reduction in persuasive performance under carefully controlled conditions. The null customization effect further identifies a boundary condition by suggesting that increasing personalization alone may not improve outcomes when message content, structure, and evidence are held constant and the intervention is brief.

Finally, we acknowledge several limitations. First, this study did not include a manipulation check that directly assessed whether participants perceived the intended distinctions among generic, demographically targeted, and psychologically tailored messages. We avoided asking explicit customization questions before the focal outcomes because such questions could have sensitized participants to the manipulation and study purpose. Nevertheless, the absence of a direct check limits interpretation of the nonsignificant effects. Future research should validate the manipulation in an independent pretest or administer manipulation-check items after the primary outcomes. Second, customized health messages may have unintentionally reduced the readability and coherence of the original professionally developed message. Future research should therefore explore alternative communication techniques, such as narrative storytelling, to enhance engagement while maintaining high message quality. Third, the characteristics of the sample may limit the generalizability of the findings. Participants were recruited through an online panel and may therefore represent individuals with relatively high levels of digital literacy, health literacy, and familiarity with online health information. Future research should examine the effectiveness of GenAI-generated customized messages among more diverse populations and in real-world healthcare settings. Finally, future research should incorporate other potential factors that may influence the effectiveness of customized messages, including health literacy, prior screening experience, cultural background, and trust in AI. These factors may shape how recipients evaluate AI-generated health information and perceive its believability. Examining them would provide a more nuanced understanding of when and for whom AI-generated health messages are most effective.

4.2. Conclusions

Viewed from a patient-education perspective, the findings indicate that neither greater customization nor GenAI generation automatically improved responses to a single, standardized educational message. For patient educators and healthcare practitioners, this shifts attention from the production source alone to the clarity, readability, coherence, naturalness, credibility, and clinical appropriateness of the information provided. GenAI may assist in drafting scalable educational materials, but professional review remains necessary to ensure accuracy, cultural appropriateness, responsiveness to patient concerns, and suitability for use alongside patient-provider communication.

4.3. Practical implications

GenAI can support the efficient and scalable development of customized health messages. However, practitioners should use high-quality, evidence-based materials as the foundation for customization and ensure that tailored content is integrated smoothly. Professional review remains important for maintaining readability, accuracy, credibility, and cultural appropriateness.

CRediT authorship contribution statement

Haoran Chu: Writing – original draft, Validation, Supervision, Project administration, Investigation, Conceptualization. **Fevi Rose Paro:** Formal analysis. **Wenshuo Zhang:** Writing – original draft. **Michelle Ruiz:** Data curation. **Hanzi He:** Methodology, Data curation, Conceptualization. **Shuo Yao:** Conceptualization.

Ethical approval and informed consent statements

This study was reviewed and approved by the Institutional Review Board of the University of Florida. All participants provided informed consent prior to participation.

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Declaration of Competing Interest

The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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